BRAC University (Department of Computer Science and Engineering)
CSE 221 (Algorithms) for Spring 2026 Semester
(NO EXTRA PAGE IS ALLOWED)
Quiz 1
SET - A

Student ID: **Full Marks: 20** **Section:**
Name: **Duration: 25 minutes**

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1. Write down the time complexity of the following code snippet in **Big-O** notation. Here, n is an input variable and k is a constant. **(5 marks)**

```
for (int i=1; i<=n; i+=k)
{
   int sum = 0;
   for (int j=n; j>=1; j=j/k)
   {
      for (int m=1; m<=n; m+=2){
         if ( m%2==0) break;
         sum = sum + 10;
      }
   }
}
```

2. For attending the quiz, you have earned **2 marks**.

3. Write down the rules/steps/assumptions for converting a growth function to asymptotic time analysis. **(3 marks)**

4. You are given an array which denotes a unimodal function with ***a plateau***. The sequence is strictly increasing, then the peak element may repeat several times (this region is called plateau), then strictly decreasing. A sample input can be like the following array:

Index	1	2	3	4	5	6	7	8	9	10	11	12
Value	3	5	6	7	8	8	8	8	8	5	2	1

(i) Draw the function and identify the plateau region. **(2 marks)**

(ii) Design a condition or property which will divide the array into two parts (First all T then F, or F then T). **(3 marks)**

(iii) Write down the code for binary search for determining the leftmost peak element in the plateau region. **(5 marks)**

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Quiz 2

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1. You have learned how the merge step of MergeSort works, where two sorted arrays are combined into one sorted array. Now, consider a slightly different case. Each element has two values: v**alue and priority**. They are given in two separate arrays: **A_Value[], A_Priority[], B_Value[], B_Priority[].** Both arrays are already sorted using the following rule:

   a) Smaller value comes first **b) If values are the same, smaller priority comes first**

   Write pseudocode/code for a function ***DoubleMerge(A_Value, A_Priority, B_Value, B_Priority)*** which merges them into **C_Value[] and C_Priority[]** so that the final arrays are also sorted using the same rule. Your algorithm must run in O(n) time. **(10 marks)**

A_Value	2	4	4	6
A_Priority	3	2	7	5

B_Value	1	4	6	8
B_Priority	5	1	2	5

C_Value	1	2	4	4	4	6	6	8
C_Priority	5	3	1	2	7	2	5	5

2. For attending the quiz, you have earned **2 marks.**
3. Show the simulation of the Partitioning step for the following array (**Pivot is 18**). **(8 marks)**

Index	1	2	3	4	5	6	7	8
Value	26	20	13	5	**18**	3	2	11

Bonus (3 marks)

Refer to Question 1. Suppose you want to use **DoubleMerge** as the **merge step of a stable MergeSort algorithm**, and you want the overall sorting to be stable. What should you store in the second array (instead of Priority) so that DoubleMerge can ensure stability? Briefly explain how this helps maintain the original order of equal elements.

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Quiz 3

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1. Write the difference between connected components and strongly connected components in a graph. **(5 marks)**

2. For attending the quiz, you have earned **2 marks.**

3. A graph always has a unique topological sorting. Do you agree or disagree with the statement? Explain your reasoning. **(3 marks)**

4. Draw a graph with 10 vertices where there will be 9 tree edges, 4 back edges, 1 cross edge and 2 forward edges. You need to show the labeling of the edges clearly. **(7 marks)**

5. How will you reconstruct a cycle after finding a back edge? **(3 marks)**

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Quiz 4

Student ID: **Full Marks: 20** **Section: 20**
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1. At a university, there are N students, numbered from 1 to N. Initially, each student belongs to their own separate club. Over time, some clubs decide to collaborate and merge into one larger club. You need to manage these club connections and answer some queries.

   Join (A, B): Student A's club and student B's club merge into one club. *If they are already in the same club, do nothing.*
   Same (A, B): Check whether student A and student B belong to the same club. Print 'Yes' if they are in the same club, 'No' otherwise.

   I. If N = 6, then write down the parent array. **(2 marks)**

   II. For the following sequence of operations, manually simulate the parent array from I after each operation using **Disjoint Set Union (DSU)** and also draw the trees accordingly: **(10 marks)**

   a. Join (1, 2)
   b. Join (2, 3)
   c. Same (1, 3)
   d. Join (4, 5)
   e. Join (3, 5)
   f. Same (6, 1)

2. For attending the quiz, you have earned **2 marks.**
3. A graph with negative edge weights will always contain a negative cycle. Do you agree with the statement or not? Provide examples or explain your reasoning. **(8 marks)**

Bonus: (2 marks)
Explain path compression with a proper example. You must show the necessary trees and parent array.